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PROSTHETICS AND ORTHOTICS



Improving access to prosthetic services in Western Nepal: a local stakeholder perspective

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ABSTRACT

Purpose: Evidence of effective strategies to improve access to assistive technology (AT) like prostheses is limited, especially in rural and remote areas of low- and middle-income countries where unmet needs are the greatest. This study aimed to identify barriers and facilitators to accessing prosthetic services in rural areas of western Nepal and explore strategies to improve access from the perspective of local stakeholders.

Method: Semi-structured interviews were conducted with 13 service providers and consumers. Barriers and facilitators of access were highlighted and potential solutions to overcome access barriers were explored using thematic analysis of transcripts.

Results: Six themes concerning barriers and facilitators arose: awareness and literacy of prosthetic services, attitudes and belief systems, financial supports, geographical access, health system and referral processes, as well as service provider capacity and regulation. Six themes regarding potential solutions were also identified: awareness campaigns, procurement pathways, referral pathways, subsidised and shared costs of AT, professional support networks and development opportunities, as well as task shifting and sharing.

Conclusions: While facilitating interventions support access to prosthetic services, they remain insufficient to overcome several barriers that continue to inhibit this access. Nevertheless, real opportunities to alleviate barriers and address the unmet need exist and must be explored.

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► IMPLICATIONS FOR REHABILITATION

- It is advised that rehabilitation professionals, particularly those delivering prosthetic services in rural and remote settings of Nepal, consider piloting alternative service delivery strategies which utilise resources accessible to them in order to overcome several existing barriers to AT access e.g., task shifting.
- Rehabilitation professionals in Nepal have the opportunity to support others working in low-resources settings to expand professional development opportunities through online and digital platforms.
- Gathering and sharing data on the current state of prosthetic and AT service delivery within Nepal is highly valuable to the development of the prosthetic rehabilitation profession and ultimately equity in access to appropriate AT.
- Proposed interventions identified within this study can potentially serve to guide stakeholders in rural and remote settings of other LMICs in developing strategies to overcome barriers to AT access suitable to their context.

Introduction

By improving mobility and independence, assistive technology (AT) can mitigate barriers to health and support services, employment, and educational opportunities [1,2]. Strengthening and extending access to AT for those who need it is thus central to inclusive socioeconomic development and optimising health outcomes for both individuals and communities [3,4].

As a signatory to the Convention on the Rights of Persons with Disabilities (CRPD) [5,6] and with the aim to meet the United Nations Sustainable Development Goals (SDGs), Nepal has demonstrated its commitment to ensure equitable access to AT [7,8], and the right to access AT is also enshrined in the Nepal

Disability Act 2017 [9]. However, most AT services operate in urban and peri-urban areas [10–12]. Like other low- and middle-income countries (LMICs), for the 80% of the population who live in remote parts of Nepal [13], access to AT is especially difficult [14,15].

In the remote districts of western Nepal, characterised by slow economic development and poor health outcomes [16,17], the rates of people living with a physical disability are particularly high. The estimated disability prevalence in the Kalikot district is twice the national average [13] yet according to a 2017 report from Nepalganj Medical College, only 14% of people living with a physical disability in Kalikot have received any rehabilitation service [18].

Table 1. Participant demographics.

Participant	Sex	Location of work	Clinical services	Service management/development	Advocacy and support
01	F	Rural/Urban	X		
02	M	Urban	X	X	
03	F	Rural	X		
04	M	Rural/Urban		X	
05	F	Rural/Urban	X		
06	M	Urban	X	X	
07	F	Rural/Urban	X		
08	M	Rural/Urban	X		
09	M	Rural/Urban			X
10	M	Urban			X
11	M	Rural			X
12	M	Rural/Urban	X	X	
13	M	Rural/Urban		X	

Access to specialised care and assistive products with unique clinical and technical requirements like limb prostheses are potentially even more challenging. Prostheses are included on both the WHO Assistive Technology Priority List [19] and the Priority Assistive Product List of Nepal [20]. Evidence concerning unmet needs for AT is generally lacking [21] and methods for addressing these needs are contested [22]. Previous studies have identified barriers and facilitators to prosthetic service provision in diverse LMICs [12,23,24]. However, there is a paucity of evidence examining solutions, particularly local-stakeholder-driven solutions, to overcoming barriers and supporting AT access [25,26].

Current discourse and efforts to strengthen access to AT often focus on new technological innovations, with prominent examples of telerehabilitation and additive printing [27]. While these have transformative potential, it is also necessary to understand whether and how they are likely to be implemented and integrated into existing structures. We therefore argue for a broader understanding of “innovation” in delivering specialised care in complex conditions that recognises and builds on promising local interventions.

Here, we examine a case study from Nepal where locally driven innovations have emerged to improve access to prosthetic and rehabilitation services. We aimed to identify current barriers and facilitating factors to accessing prosthetic services in rural areas of western Nepal and explore strategies to improve access from the perspective of local stakeholders. While these concepts characterise and are designed for the unique geographical, economic, and cultural landscape of remote regions in western Nepal, specifically focusing on the Kalikot district, they provide an opportunity to develop illustrative theories which may generalise these specific experiences to other similar contexts.

Materials and methods

Study design

This qualitative study based on semi-structured key informant interviews (KIs) was conducted through field research by visiting international collaborators. Kalikot in western Nepal was selected as a focus because of its disproportionately high rates of physical disability and observed local innovations to tackle barriers to access.

Recruitment of participants

Purposive sampling was used to invite eligible adult participants to take part in KIs. Participants were identified by the research team as representatives from key stakeholder bodies involved in prosthetic service provision and/or advocacy for people living

with a physical disability in Nepal, including those from Kalikot. Participant characteristics are described in Table 1.

Potential study participants were provided with a Plain Language Statement in both English and Nepali detailing the purpose of the study and processes in data collection, storage and analysis. They were then offered the opportunity to voluntarily sign consent forms in either English or Nepali if they wished to participate in the study.

Data collection

Using a question guide developed by the authors (MF, WP), KIs were conducted either in person at the participant's workplace (10 interviews) or *via* online video conferencing platform LINE (LINE Corporation, Tokyo) when in-person interviews were not possible (3 interviews). Over-arching, broad questions consistent with the study aims were initially developed and refined by the researchers [28]. The interviewer (MF) used the question guide as a broad structure, but used follow-up, probing questions to clarify responses or deepen enquiry. The KIs were conducted in English (8 interviews) or in Nepali with an interpreter if the participant or interviewer requested assistance in translation between English and Nepali (5 interviews). The interpreter was certified by the National Accreditation Authority for Translators and Interpreters (Australia) and was included in the interview *via* LINE. All KIs were audio recorded and files were saved on a secure cloud-based file-sharing platform.

Once KIs were completed, audio recordings were transcribed verbatim. One researcher (MF) made a preliminary review and basic coding to determine the need for further exploration. After the initial cohort of interviews, it was determined that data saturation had been reached and that no further KIs were required.

Analysis

Transcribed KIs were entered into qualitative data analysis software NVivo (QSR International, Melbourne, Australia). A code template was developed based on high level themes of facilitators, barriers, or strategies to improving access to prosthetic services for people living with an amputation in Kalikot. This template was developed by authors (MF, WP) with an inductive approach and was refined with further collaboration and consensus. Within each were code labels determined as possible factors relating to either availability of service provision (supply) or level of activity to acquire prosthetic services (demand). Data were then coded using this template. Additional ideas that emerged were added to the codes.

Codes were then categorised and analysed to identify themes pertaining to facilitators and barriers to prosthetic service access

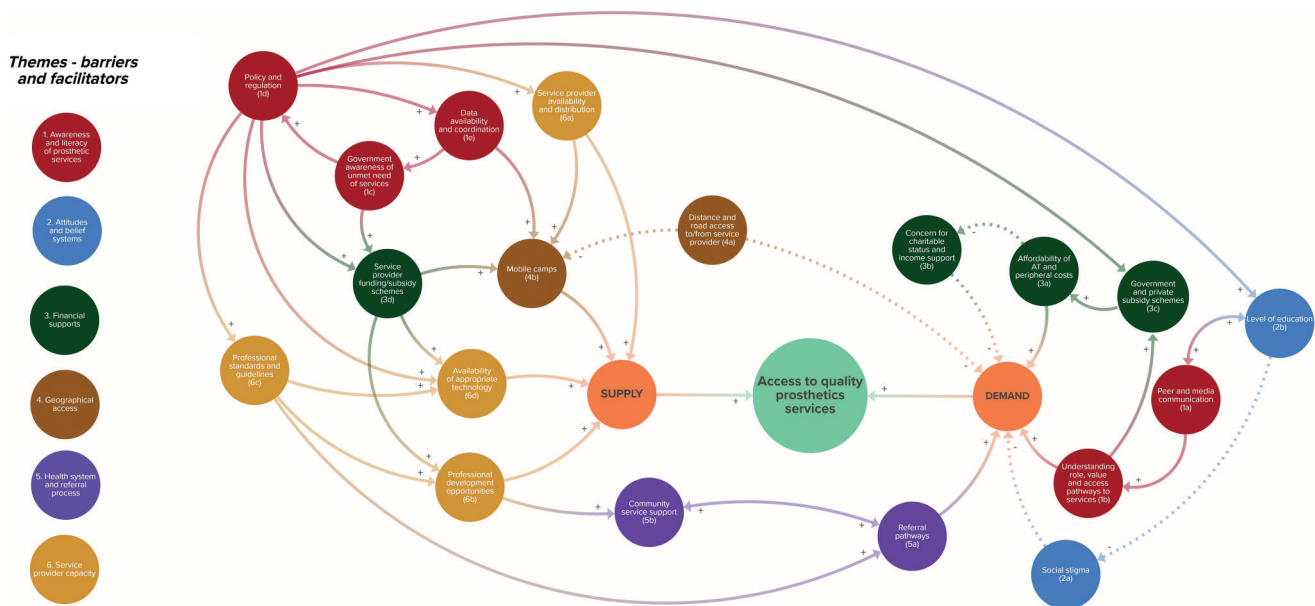


Figure 1. Interrelationships between influencing factors, or subthemes of both supply and demand components of prosthetic service access in rural western Nepal. Subthemes fall within themes – barriers and facilitators (left) of corresponding colours and numerical reference e.g., the subtheme coloured green and with reference (3b) fall under the theme coloured green and with the numerical reference (3). Solid lines: positive relationship; dashed lines: negative relationship.

as well as potential solutions to overcoming barriers and improving access. These themes and their interrelationships were then corroborated by referencing back to the summaries of data and overarching succinct phrases to underpin those themes. Themes were determined individually by authors (MF, WP) before being compared, discussed and reconciled until a consensus was achieved.

Initial findings and interpretations from the data in the form of themes were then presented back to two groups of study participants in analysis workshops for further clarification, interpretation, and further development. Themes were presented sequentially using PowerPoint slides, after which, the group was invited to discuss whether they agreed with the findings and how potential solutions could be applied in a practical way. Open discussions also included evaluation of solutions as to how difficult they would be to implement and how impactful their implementation could be to improving access.

Ethics

Ethics approval to conduct this study was provided by the University of Melbourne's Faculty of Medicine and Dentistry Human Ethics Sub-Committee (Ethics ID 1852771).

Results

Table 1 describes participants. Thirteen (4F:9M) participants took part in KIs. Three participants worked in advocacy and support, while the rest worked in a combination of clinical and service management roles.

Themes – barriers and facilitators

Six themes concerning barriers to accessing services and local solutions emerged from stakeholder testimony and analysis workshops: awareness and literacy of prosthetic services, attitudes and beliefs, financial supports, geographical access, health systems and referral processes, and service provider capacity. These

barriers and facilitators form a complex set of interrelationships which collectively determine the level of access. **Figure 1** illustrates the overall barriers and facilitators and links between them.

Awareness and literacy of prosthetic services

Respondents suggested that lack of awareness and literacy of prosthetic services among consumers and health workers is a major barrier to demand (right hand side of **Figure 1**). Despite activities promoting services such as through peer communication, media and Disabled Persons Organisations (DPOs) (1a), participants commonly reported low community awareness of the role, value and access pathways to prosthetic services (1b), including through government support schemes such as Disability ID Cards. The result has been reduced demand for primary and follow-up services. Respondents indicated this arises from a mix of limited community knowledge and limited communication from professionals:

Maybe because of the education level, so many of them do not know that they can get these devices. Lack of communication, lack of work done by the health workers, lack of penetration to that level. (M, Rural/Urban, Management/Development)

How they see us is like a shoe-maker... they say 'oh I don't have time for follow-up', but they have time to see the doctor. (M, Urban, Clinical, Management/Development)

Respondents described several limitations to the awareness of the unmet need at different levels of government (1c) that result in barriers to supply of access (left hand side of **Figure 1**). Direct government support was acknowledged, with modest travel subsidies for persons with a Disability ID Card noted as a facilitator of demand among those with access to the scheme, as was local municipality assistance in conducting mobile rehabilitation camps. However, a lack of public service provision, supportive policy and regulatory frameworks for the private sector were prominent barriers to supply (1d). It was suggested that these challenges may arise from an observed lack in general understanding of the overall need and value of prosthetic services.

Many stakeholders and also governments are not really aware of our services. For the physio they are already aware about the physio

services and then they have vacancies in government hospitals. But for prosthetics and orthotics this is still challenging that people are still not aware of our services. (F, Urban/Rural, Clinical)

This was also said to be influenced by insufficient coordination of data between service providers and support organisations, partially due to a lack of willingness for it to be shared between facilities (1e). This has resulted in difficulties in prioritising service delivery and increasing government awareness.

They say that 'we have dealt with 500 patients'... But how many of them actually get benefit? Nobody counts on that. (M, Urban, Advocacy, and Support)

Attitudes and belief systems

Respondents described social stigmatisation of people living with a disability, particularly amongst older and less educated populations leading to prejudicial deprioritising of disability issues generally and access to AT specifically (2a). One respondent commented that *"people with disability are considered a burden for the family, burden for the society, they do not have any role in economic empowerment, they cannot be a bread-winner for the family,"* but noted the changing context and opportunities shifting attitudes around disability with higher levels of education and understanding of the rights of persons with a disability (2b): *"...however, it is changing because mostly educated people or people who have got knowledge about these things have got slightly different views"* (M, Rural/Urban, Advocacy and Support).

Financial supports

According to respondents, the typically low socioeconomic status of most people living with an amputation in Kalikot inhibits the ability to meet direct and indirect costs of AT (e.g., transport, lodging, loss of income for individual and support network) associated with accessing care (3a). Further, while no specific examples were given, some participants suggested potential prosthesis users might avoid uptake of care for fear they would be less likely to receive charity which they depend on for financial support (3b). Consequently, they may under-utilise or reject available services.

priority is much less towards the prosthetics and it is much more the living... They just bother 'how I should live, what I should earn and what I should eat, what I should wear.' (M, Rural/Urban, Clinical, Management/Development)

they don't want a prosthesis because they want to live so that they get sympathy from the people as well as allowances from the government. (M, Rural/Urban, Management/Development)

Means-tested government and private support schemes such as Disability ID Cards and external funding support for mobile rehabilitation camps lobbied from local government and DPOs are available to assist in mitigating the financial barriers to accessing prosthetic services (3c). Recipients of services delivered through mobile rehabilitation camps will commonly incur costs commensurate to their wealth and assets. The vast majority of those in rural and remote locations in western Nepal are regarded as of the lowest socioeconomic bracket and as such are commonly offered services at no cost.

However, the effectiveness of these subsidised services is dependent on community awareness of such supports and the capacity to acquire them. The availability of necessary external funding support also relies on continuous lobbying from service providers and advocates and was described as limited and insufficient to supply the unmet need with an appropriate, timely service.

Further, while Disability ID Cards can provide discounts on travel and primary healthcare costs relative to a classification of disability, they do not subsidise any direct costs of AT nor all the indirect costs of acquiring services outside mobile rehabilitation camps. Participants reported that these costs deter people with amputation from accessing prosthetic services from the centralised provider when required.

Last year we had one camp, at that time, some of them miss those because of the limited amount of funding. This year I am hopeful that the government will give some funds... there are so many missed ones, so many missed people who have not been able to get prostheses. (M, Rural/Urban, Management/Development)

While predominantly privatised, service providers also depend on funding support from external sources to supplement the limited income they receive from service users (3d). Currently, availability of funding support was described as inadequate for service providers to ensure professional development opportunities for staff, a high quality of products and resources as well as appropriate distribution of staff, especially for the purposes of conducting mobile rehabilitation camps.

...in general situations for remote regions our camps are conducted every three years I would say, because it involves a lot of cost for the centre to do those mobile camps. For example, the entire rehab team has to go to the area, they have to take all of their workshop and their materials, components with them. It's a huge cost. (F, Rural/Urban, Clinical)

Geographical access

Long distances, limited road access between communities and the point of service, limited transportation options, and difficult terrain to traverse by foot were identified as key barriers to both supply and demand for services (4a). This is especially relevant for those living with a lower limb amputation and without a suitable or functional prosthesis.

The roads are only up to some select locations. The people away from those locations, it is very difficult for them to come. Carrying the people with disabilities to that location is a big challenge. (M, Rural/Urban, Clinical, Management/Development)

As alluded to above, one local solution has been mobile rehabilitation camps in which central service providers travel to rural and remote areas to provide services (4b), with support from local government and/or non-government organisations (NGOs) such as Humanity and Inclusion. However, these mobile services are reportedly infrequent, due to factors including inadequate financial assistance and limited service provider capacity.

Health system and referral process

Weak and fragmented referral pathways limit access to available care and the quality of follow up (5a). Community Disability Workers (CDWs) have been employed across multiple districts in western Nepal to provide links between tertiary and community rehabilitation care as well as instruction for people living with a disability on physical exercises (5b). While available, community-based services are not used effectively through post-amputation referrals. Many new amputees are lost to follow-up as they are not supported to access management before or immediately after amputation.

Some of them go back to their place, their home, town or village, then after one or two years, some community disability workers catch them, they counselled them, then they come into contact. (M, Rural/Urban, Management/Development)

Service provider capacity

Multiple rehabilitation professionals including physiotherapists, doctors, CDWs, and prosthetist/orthotists, are all able to attend

mobile rehabilitation camps. However, they are limited in their availability and distribution, specialised skills, and in geographical reach (6a). These factors combine to inhibit service quality and continuity. Finite human and financial resources constrain innovative solutions like camps: One clinician reflected: *"when we have to move to a camp, then our centre will be closed. We do not have enough staff I think"* (F, Rural, Clinical). Mobile camps are therefore infrequent, as described above. CDWs were regarded as having limited knowledge of AT and prosthetic management, with only three months of physical therapy training. At the time KIs were being conducted, only one International Society for Prosthetics and Orthotics (ISPO) associate prosthetist/orthotist and one CDW were employed to provide services across the entire former mid-western development region.

The limited skill set of health professionals was considered a by-product of limited local opportunities for professional development (6b). Workshops and online courses were recognised, though were considered too infrequent and do not provide enough practical content to build the technical capacity of the health profession.

The education level, like with the skill of fabrication, and the knowledge is also stagnant. We don't have any courses, continuous professional development. I have completed a bachelors, and that's it. (M, Urban, Clinical, Management/Development)

Compounding a lack of training is a lack of professional standards, guidelines, and policies (6c). Without regulated practice, the opportunity for unqualified or insufficiently trained practitioners to emerge was described as *"very dangerous for the service"* (M, Urban, Clinical, Management/Development).

A unique characteristic of AT and prosthetics specifically is the need for a combination of "mass market" and specialised raw materials and components. Unreliable procurement pathways combined with limited human resources threaten both the scope and quality of services. Without a local distributor of materials and components, service providers experience significant challenges in issuing affordable, high-quality technology appropriate for the environment of use in a timely manner (6d).

Sometimes it will be 4/5 months to get the components. So if we are planning procurement, we need to start planning 6 months before thinking that we need components, so it's quite difficult. Also, we have problem with customs, so we need to pay for the customs and VAT also. (F, Rural/Urban, Clinical)

To manage the difficulty of long procurement times and high costs, goods are ordered in bulk, most commonly from India or China. This however, limits the opportunity for all service users to be fitted with components which are best suited to their presentation and needs.

Themes – solutions

Six themes concerning solutions or potential strategies to improve access to prosthetic and rehabilitation services were identified. These solutions were derived from the "explanatory mechanisms" of the barriers and facilitators identified through expert stakeholder testimony and revised through further consultations. Each theme includes details of potential interventions and the rationale, perceived difficulty, and anticipated impact.

Awareness campaigns, strategies, and resources

Stakeholders supported the conclusion that improving the awareness of the unmet needs and value of prosthetic services among all levels of government would indirectly support efforts to strengthen access. Collective efforts of service providers, DPOs,

and professionals to campaign for supportive public policy require quality evidence, as well as recommendations of enforceable standards and guidelines for the P&O profession. The most important policies were for financial support or incentives for the sector (e.g., tax exemptions for imported materials and components), and for increased subsidies for AT user costs and peripheral expenses. Strengthening regulation of the sector through co-development of standards and guidelines for service providers, relevant to allocation of human and material resources was also strongly indicated.

The main strategy to expand the service is, first of all, the government should think about the policy and the standard and the guidelines to how the service could be provided from the ground level, from the community level. (M, Urban, Clinical, Management/Development)

Participants also viewed service providers, DPOs, and community health personnel to be responsible for engaging community with educational resources promoting pathways to service accessibility, realistic expectations of services, as well as the rights of people with a living with a disability. Programmes involving visual resources as well as peer support networks, particularly for those who are illiterate, could be implemented on a non-government level with little requirement for funding support. Such programmes, however, were said to be dependent on geographical access and willingness of the appropriate community members.

Accurate data on the unmet needs of people living with an amputation were regarded as a major requisite to guide and enact legislative change, endorse appropriate funding allocations, guide service delivery strategies and prioritisation. Mobilising NGOs and community workers to collect appropriate, objective, and accurate data in the field *via* robust collection and analysis methods were strongly recommended. Sharing and collating of these data between relevant bodies was also recommended.

Procurement pathways

Overcoming the identified financial and logistical challenges in procuring technology appropriate to the user environment was described as essential to supplying high quality prosthetic services that are safe, effective, and durable. Procurement pathways alternate to current practices were discussed; however as above, lobbying for policy change regarding taxation of imported components and materials *via* the existing procurement pathways was most commonly considered the most feasible strategy.

Other strategies such as in-country distribution of components *via* a centrally located bank of former demonstration components from international sources were met with scepticism. Participants described anticipated barriers of high importation costs, discontinuity of supply, and inequity of distribution between states and districts. For such a concept to have potential for long-standing success, professional standards, and quality control would need to be established and enforced, while exemptions for importation taxes must also be issued.

Not having the legal framework to establish such an organisation. Now we're not even able to get a tax offset on the components that we get in Nepal. Thinking about making a central bank and establishing the bank, I don't think it is very possible in the near future. (M, Rural/Urban, Clinical)

Referral pathways

Bidirectional referral pathways between community or district level health posts and larger central rehabilitation centres were also discussed as a measure to minimise the number of potential prosthetic users lost to follow-up and improve the level of demand. A model previously unexplored, participants suggested

this could only be successfully implemented if the requirements of people living with an amputation and where services can be offered were clearly understood by local health services e.g., CDWs. Establishment of secure data collection and storage methods to ensure safe and accurate transfer of information through referrals was also deemed necessary.

We have to coordinate with the local people there, know exactly how many clients are there, people there who are in need of those kind of things, or who needs rehabilitation, who needs prostheses. (M, Rural/Urban, Management/Development)

Subsidised and shared costs of at and peripheral expenses

To overcome financial barriers to access, further lobbying for continuous funding support from district and local level governments was strongly suggested, especially given the deconcentration of funding, authority, and responsibility from the federal-level government under the new recently written constitution. This required dissemination of data to be able to quantify the level of funding required to cover direct and indirect costs of the service and, again, highlight the awareness of the unmet need.

We can collaborate with the DPOs or other organisations to provide some support for the transportation as well as food when they come for the camp as well. (F, Rural, Clinical)

One participant identified income generation support and advocacy for employment as a strategy to increase income and improve agency in accessing appropriate care. However, others regarded such a strategy to be unlikely to directly influence service demand due to prioritisation of other living expenses.

Professional support networks and development opportunities

To improve quality and supply of services, professional development of service providers through further in-house workshops was raised. Workshops were considered highly impactful and valuable in exposing service providers to hands-on training. To do so, identified challenges included recruiting enough participants from both urban and regional areas to make training viable.

An online mentorship programme, facilitated by senior local Nepali service providers and international support was considered more feasible. Such a programme would allow for continuity in professional support relevant to the existing case load of Nepali service providers while avoiding geographical and financial limitations. However, this requires willingness of potential suitable mentors to maintain commitment to such a programme as well as sound internet connectivity.

Task shifting and sharing

The concept of task shifting was explored in KIIs as a strategy to alleviate several barriers of both the demand and supply components of access. In combination with a bidirectional referral pathway strategy, as described above, building the capacity of local human resources such as CDWs to perform minor prosthetic services (e.g., replacing a broken foot or issuing stump socks) was described as a difficult intervention to implement, though highly impactful as a concept. Considerations for such a strategy included the requirement for established standards and guidelines defining the roles and responsibilities of the community level service, certifiable training delivered by higher-level service providers, monitoring and evaluation of community service, clear communication pathways between levels of service, as well as selective recruitment of competent community-level workers.

This idea is very good for the community level actually. This is the ideal way to do that, to reach the services in that community. But what I was saying before, until and unless the Nepal government don't have any good strategy, guideline, standard and policy, that is very dangerous for the service. (M, Urban, Clinical, Management/Development)

Discussion

This study explored barriers of access to prosthetic services for people living with an amputation in rural and remote districts in the west of Nepal as well as existing local interventions in place to alleviate these limitations. Expert stakeholders identified overarching themes of awareness and literacy of prosthetic services, attitudes and belief systems, financial supports, geographical access, health system and referral processes, as well as service provider capacity and regulation as major overarching themes of barriers and facilitators. Illustrative and potentially generalisable solutions to barriers were co-developed between researchers and stakeholders to build on existing strategies in locally relevant ways.

Despite local innovations and commitment to facilitate prosthetic service access, best exemplified by partnerships between local government agencies and private service providers to support mobile rehabilitation camps, unmet needs in rural populations remain high. Access in this context is determined by variables such as transport opportunities and referral processes and a series of social structures such as socioeconomic status. These conditions reinforce inequities and constrain economic participation, social inclusion and improved health outcomes.

Many of the barriers to the supply and demand of access to prosthetic services identified as themes in this study echo those found in other LMICs [12,23,29–31]. Several of the same barriers have also been identified to inhibit access to primary healthcare in rural and remote communities of western Nepal [32,33]. Given the consistency of such observations across several different contexts, it is unsurprising that these barriers also apply to the specific requirements of prosthetic and physical rehabilitation services for people living with amputations in and around Kalikot.

Resolving these inequities requires solutions at multiple levels of the health system and from diverse stakeholder groups. WHO has made recommendations for policy development to promote prosthetic service access in LMICs [34]. These include regulated standards and guidelines for all levels of service provision, a commitment to sustainable public and private financing models, material importation expense exemptions and establishment of data collection and management protocols to monitor and evaluate the quality, efficiency, and effectiveness of interventions [11]. Their consistency with findings in Nepal illustrates the current gap between normative frameworks and recommendation and the necessary resources and incentives to act on them in diverse contexts. Barriers to their implementation are possibly shared by the complex and interwoven barriers to service access which they are aimed to alleviate.

Improved awareness of the unmet need, access pathways, and value of prosthetic and rehabilitation services among duty-bearers are necessary to lobby for policies that support both the demand and supply of prosthetic service access. This depends on availability and dissemination of accurate and appropriate data to justify policy changes in accordance with the mandates of the CRPD, SDGs and Nepal Disability Act 2017. Current efforts to quantify unmet needs for AT including the WHO's rapid Assistive Technology Assessment (r-ATA) survey [35] and Global Report on AT [36] provide the necessary method, with the former recently being implemented within Nepal. Furthermore, a rehabilitation

module has now been integrated into Nepal's centralised Health Management Information System for monitoring and evaluating the performance of rehabilitation services [37]. While valuable and identified as an important strategy, continuous collection, sharing, and dissemination of such data, for purposes of advocacy, depends on strong political will and allocation of scarce health resources in a highly competitive environment.

Using existing resources in Nepal and identifying those necessary but not yet available is crucial to the implementation of strategies required for effective collection of data as well as promotion of services. Private providers supply a large majority of existing services throughout Nepal and evidence from this study highlights their centrality to the local arrangements that have evolved to develop and expand the reach of services. The development of further solutions to improving access is thus likely to continue involving the private sector.

Without a fully subsidised public service and with large barriers to self-funding private services, alternate financing models, such as those used to support mobile rehabilitation camps, must be further developed to a more continuous and inclusive arrangement whereby services may be available on an individual as-needs basis. Many successful innovations in private health and AT service delivery in LMICs have been initiated by entrepreneurial capital support and rigorous operating expense management, such as engaging lower cost community-based service providers [38,39], a strategy also recommended by participants of this study. Task shifting with community health workers as part of community-based rehabilitation (CBR) programmes in other contexts has also expanded the reach and availability of health and AT services, reduced inequalities between population groups, gender-based barriers to accessing health services, and out-of-pocket expenditure on peripheral costs [38,40–44].

WHO and ISPO have provided guidelines on CBR implementation in prosthetics and orthotics [34,45,46], including the establishment of a basic level of service administered by community workers. However, their application is patchy. Given the presence of these guidelines, the success of CBR in other contexts as well as the projected positive impact proposed by participants of this study, several supply and demand barriers including those of limited service provider capacity, financial constraints and geographical barriers could be mitigated by adopting such an approach [46]. This could involve expanding the role of CDWs beyond physical therapy education to providing minor technical supports as a referred service and collecting more extensive data. Yet, it must address challenges pertaining to financial, staffing and resource capacity, professional development, communication channels, and the establishment of regulated policy defining respective roles, all of which are likely to have inhibited application of these guidelines elsewhere [10,29,47]. Integration of rehabilitation services into primary level healthcare in Nepal can also be supported with the guidance of documents such as the WHO Basic Rehabilitation Package [37,48].

Low-cost remote services may also be made possible through tele-rehabilitation practices. Participants endorsed the concept of online mentorship for the purpose of expanding professional development opportunities. Pending sound internet connectivity and willingness of potential mentors, such a model could be established similar to Project ECHO® (University of New Mexico, Albuquerque, NM) [49–51] or the Motivation Australia Pacific Wayfinders programme operating in Pacific Islands countries, whereby local health professionals within the community are supported by larger centralised services *via* a video conferencing platform [52]. Such programmes have been demonstrated as a

cost-effective model to increase the reach of service delivery and support professional development of service providers at variable levels of the health service, including the community level [50,53]. Implementing this strategy could be strongly influential in overcoming limited service provider capacity and allowing Nepal to realise the critical mass of service providers necessary to improve the supply component of access.

Efforts to overcome difficulties regarding procurement of environmentally and culturally appropriate prosthetic materials and components have been explored in other contexts where similar challenges exist. Several attempts have been made to introduce low-cost, durable, locally manufactured AT [51,54–56], yet poor outcomes have been attributed to inadequacies in the construction, material, socket fit, and training of personnel [57–60]. In other LMIC environments, however, local manufacture of AT is commonly practiced and with a high degree of success [38]. Manufacturing assistive devices in Nepal at scale are yet to be realized [61] though there are some initiatives in place exploring the role of 3D printing to overcome these challenges [27,62]. Though not explored in depth in this study, learnings from previous contexts should be considered alongside guiding documents in AT procurement [63] for strategies of local manufacturing to be effective in improving the level of access for people living in remote settings. Nevertheless, the findings from this study concerning professional standards of service should also be applied to products that are to be procured or manufactured, while the feasibility of task shifting programmes is dependent on the efficient and sufficient supply of procured materials to the community level [63].

Strengths and limitations

A key strength of this study is its situation within locally led programmes and giving voice to the experiences and expertise of local stakeholders. Rather than implementing and evaluating new and entirely externally driven approaches, this study explored challenges and solutions from an inside perspective, providing new information about the real conditions in western Nepal. This introduces limitations on how the findings can be applied in other contexts. Further, while a certified translator supported KIs and without the KI question guide being initially piloted, local nuances and cultural norms cannot may not have been translated and understood completely. Perhaps most importantly, service users were underrepresented, creating opportunities for further exploration of user insights and experiences.

It must also be acknowledged that all data were collected prior to the COVID-19 pandemic presenting an influence on access to AT in Nepal. Additional barriers are likely to have been introduced over the course of 2020–2021 [64,65].

Conclusions

Although this study identified a number of interventions, which were described to facilitate access to prosthetic services in western Nepal, several barriers exist and continue to constrain this access short of what is required to meet the need. Many of these barriers pertain to the awareness of prosthetic services, geographical challenges, availability of financial supports, and the capacity of prosthetic service providers to provide an appropriate quality service.

Nevertheless, opportunities to overcome many of these ongoing barriers and ultimately improve equity of access to prosthetic services also exist. Most potential strategies could be

implemented through utilising local resources including task shifting to a community-level service, yet some would also likely require external supports such as professional support networks. With endorsement from local experts, these strategies should be explored further to determine their effectiveness in strengthening access to AT for people who require it most.

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Author contribution

Conceptualisation, M.F. and W.P.; methodology, W.P.; formal analysis, M.F. and W.P.; investigation, M.F.; data curation, M.F.; writing – original draft preparation, M.F.; writing – review and editing, M.F., Y.N., and W.P.; supervision, W.P. All authors have read and agreed to the published version of the manuscript.

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